Part A: Order of Operations

Evaluate each expression completely.

1. $(-9+(-2))^2 \div (-11)$

2. $(5+3)^2 \div (-2)$

3. $(-5)^2 - (-20) \cdot 6$

4. $(-4+12)^2 \div 2$

5. $(9+3)^2 \div (-8)$

6. $(2+14)^2 \div (-8)$

7. $18 - 8 \cdot ((-4)+18)$

8. $8^2 - 4 \cdot 2$

9. $(-2+15)^2 \div 1$

10. $24 - 4 \cdot ((-9)+24)$

11. $31 - 8 \cdot ((-5)+31)$

12. $14 - 3 \cdot ((-2)+14)$

13. $8^2 - 3 \cdot 4$

14. $40 - 4 \cdot (5+40)$

15. $(-7)^3 - 17 \cdot 9$

Part B: Negative Number Operations

Evaluate each expression completely.

1. $-12 \cdot (-4)$

2. $-9 \cdot 11$

3. $8 \div (-4)$

4. $-64 \div 8$

5. $-20 \div 10$

6. $-60 \div 12$

7. $-34 + (-9) \cdot 15$

8. $37 + 8 \cdot 6$

9. $11 + (-11) \cdot 6$

10. $196 \div 14 - (-32)$

11. $28 + (-15) \cdot (-14)$

12. $-27 + 9 \cdot (-10)$

13. $25 \div 5 - (-5)$

14. $72 \div (-6) - (-20)$

15. $33 \div 3 - 40$

Part C: Solving Linear Equations

Solve each equation for x.

1. $9x - 10 = -\frac{13}{4}$

2. $12x + 12 = -54$

3. $4x + 4 = -4$

4. $9x - 12 = 6$

5. $5x - 8 = -3$

6. $8(8x - 11) + 3 = -213$

7. $5(4x + 9) + 4 = -191$

8. $8(x - 4) + 5 = 5$

9. $3(2x + 3) - 10 = -25$

10. $8(7x - 7) - 1 = 167$

11. $-6x + 14 = 11x - \frac{23}{2}$

12. $-8x + 13 = 15x + 151$

13. $-10x - 10 = -15x + 45$

14. $7x + 4 = -3x + 44$

15. $3x + 7 = 2x + 17$

Part D: Linear Inequalities

Solve each inequality for x. Remember to reverse the inequality sign when you multiply or divide by a negative number.

1. $x - 6 > -7$

2. $5x \leq 35$

3. $x + 9 < 1$

4. $2x - 3 < -21$

5. $10x - 10 > -100$

6. $8x - 8 \leq 80$

7. $-2x + 1 \geq 21$

8. $2x + 9 > -9$

9. $-12x + 7 < -125$

10. $6x - 2 \leq 7x - 1$

11. $-9x + 13 < 4x + 65$

12. $-13x - 3 > -15x + 7$

13. $4x - 1 \leq \frac{1}{2}x - \frac{9}{2}$

14. $-2x - 12 < -\frac{5}{3}x - 10$

15. $\frac{5}{3}x + 10 < 3x + \frac{82}{3}$

Part E: Slope-Intercept Form

Rewrite each equation in slope-intercept form if needed, then identify the slope m and the y-intercept b.

1. $y = 8x$
2. $y = \frac{9}{2}x + 4$
3. $y = 3x - 5$
4. $y = 7x + 1$
5. $y = 4x - 9$
6. $y = 7x - 2$
7. $y = 9x - 5$
8. $-27x + 3y = -12$
9. $6y = 15x - 6$
10. $4y = 36x - 16$
11. $4y = 8x - 16$
12. $36x - 4y = -28$
13. $3y = 18x - 6$
14. $-9x + 3y = 24$
15. $4y = 14x - 24$

Part F: Exponent Rules

Simplify each expression. Write your answer using only positive exponents.

1. $4x^8 \cdot 9x^9$
2. $3x^2 \cdot 6x^6$
3. $9x^3 \cdot 2x^2$

4. $\dfrac{5x^6}{x^2}$

5. $\dfrac{6x^6}{x^2}$

6. $\dfrac{2x^7}{x^3}$

7. $(2x^7)^2$

8. $(4x^2)^4$

9. $(6x^2)^2$

10. $7 \cdot (3^2)^{-2}$

11. $\dfrac{5}{3^{-1}}$

12. $9 \cdot 7^{-2}$

13. $(2x^{-3})^3 \cdot x^{16}$

14. $(2x^3y^3)^2 \cdot 2x^2y^8$

15. $(6x^8y^{10})^3 \cdot 6x^{12}y^8$

Part G: Square and Cube Roots

Simplify each root completely. If the radicand is not a perfect square or cube, use prime factorization to pull out every factor you can.

1. $\sqrt{49}$

2. $\sqrt{64}$

3. $\sqrt{9}$

4. $\sqrt[3]{27}$

5. $\sqrt[3]{8}$

6. $\sqrt{216}$

7. $\sqrt{188}$

8. $\sqrt{60}$

9. $\sqrt{168}$

10. $\sqrt{204}$

11. $\sqrt{80}$

12. $\sqrt{148}$

13. $\sqrt[3]{240}$

14. $\sqrt[3]{176}$

15. $\sqrt[3]{80}$

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Simplify each expression. Write your answer using only positive exponents.

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3. $9x^3 \cdot 2x^2$

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5. $\dfrac{6x^6}{x^2}$

6. $\dfrac{2x^7}{x^3}$

7. $(2x^7)^2$

8. $(4x^2)^4$

9. $(6x^2)^2$

10. $7 \cdot (3^2)^{-2}$

11. $\dfrac{5}{3^{-1}}$

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10. $\sqrt{204}$

11. $\sqrt{80}$

12. $\sqrt{148}$

13. $\sqrt[3]{240}$

14. $\sqrt[3]{176}$

15. $\sqrt[3]{80}$

Answer Key

Part A: Order of Operations

1. -11
2. -32
3. 145
4. 32
5. -18
6. -32
7. -94
8. 56
9. 169
10. -36
11. -177
12. -22
13. 52
14. -140
15. -496

Part B: Negative Number Operations

1. 48
2. -99
3. -2
4. -8
5. -2
6. -5
7. -169
8. 85
9. -55
10. 46
11. 238
12. -117
13. 10
14. 8
15. -29

Part C: Solving Linear Equations

1. $x = \frac{3}{4}$
2. $x = -\frac{11}{2}$
3. $x = -2$
4. $x = 2$
5. $x = 1$
6. $x = -2$
7. $x = -12$
8. $x = 4$
9. $x = -4$
10. $x = 4$
11. $x = \frac{3}{2}$
12. $x = -6$
13. $x = 11$
14. $x = 4$
15. $x = 10$

Part D: Linear Inequalities

1. $x > -1$
2. $x \leq 7$
3. $x < -8$
4. $x < -9$
5. $x > -9$
6. $x \leq 11$
7. $x \leq -10$
8. $x > -9$
9. $x > 11$
10. $x \geq -1$
11. $x > -4$
12. $x > 5$
13. $x \leq -1$
14. $x > -6$
15. $x > -13$

Part E: Slope-Intercept Form

1. $m = 8, \quad b = 0$
2. $m = \frac{9}{2}, \quad b = 4$
3. $m = 3, \quad b = -5$
4. $m = 7, \quad b = 1$
5. $m = 4, \quad b = -9$
6. $m = 7, \quad b = -2$
7. $m = 9, \quad b = -5$
8. $m = 9, \quad b = -4$
9. $m = \frac{5}{2}, \quad b = -1$
10. $m = 9, \quad b = -4$
11. $m = 2, \quad b = -4$
12. $m = 9, \quad b = 7$
13. $m = 6, \quad b = -2$
14. $m = 3, \quad b = 8$
15. $m = \frac{7}{2}, \quad b = -6$

Part F: Exponent Rules

1. $36x^{17}$
2. $18x^{8}$
3. $18x^{5}$
4. $5x^{4}$
5. $6x^{4}$
6. $2x^{4}$
7. $4x^{14}$
8. $256x^{8}$
9. $36x^{4}$
10. $\frac{7}{81}$
11. 15
12. $\frac{9}{49}$
13. $8x^{7}$
14. $8x^{8}y^{14}$
15. $1296x^{36}y^{38}$

Part G: Square and Cube Roots

1. 7
2. 8
3. 3
4. 3
5. 2
6. $6\sqrt{6}$
7. $2\sqrt{47}$
8. $2\sqrt{15}$
9. $2\sqrt{42}$
10. $2\sqrt{51}$
11. $4\sqrt{5}$
12. $2\sqrt{37}$
13. $2\sqrt[3]{30}$
14. $2\sqrt[3]{22}$
15. $2\sqrt[3]{10}$